

Responses to the Editor

Manuscript ID: IPM-D-25-06425

Manuscript Title: *ALD²: Adaptive Layer-wise Denoising Decoding for Hallucinations Mitigation in Large Vision-Language Models*

Dear Editor-in-Chief,

We would like to thank you for the time and effort devoted to the preliminary evaluation of our manuscript. We carefully considered the editorial comments and have substantially revised the manuscript to improve its clarity, structure, and overall quality.

In particular, we have strengthened the abstract, clarified the research objectives, and added an explicit discussion of the theoretical and practical implications of our work. All revisions have been highlighted below.

Sincerely,
Authors

Editor-in-Chief (comments and responses)

Comment: Your abstract does not highlight the specifics of your research or findings but contains too much background information. Some details of your research would be nice, for example, numbers addressing the sample, data, percentage improvement, etc. An abstract with some details helps show the impact of your research. This does not mean making the abstract longer! – Shorter is better! Remove some of the background material and add some details of your research.

Response: We have streamlined the abstract by reducing background-related descriptions and focusing more on the key findings and contributions of our study. The revised version is provided below.

Abstract

Large vision-language models (LVLMs) have shown strong reasoning capabilities by leveraging Large language models (LLMs). However, hallucination remains a challenge for many LVLMs. Through our experiments, we identify that existing noise in shallow layers often leads to unstable token predictions during reasoning. We find that visual token pruning technologies along with multiplicative decoding effectively mitigate this interference. Building on this insight, we propose Adaptive Layer-wise Denoising Decoding (ALD²), a novel framework that mitigates hallucinations within LVLMs by dynamically selecting an optimal shallow layer for visual-token pruning and multiplicatively fusing the denoised distributions with that of the final layer. Extensive experiments on several benchmarks (CHAIR with 500 images; POPE with 500 images; MME with 2,374 cases) demonstrate that ALD² consistently improves the accuracy across multiple LVLMs. For example, we achieve an average accuracy improvement of 5.61% on POPE metrics using the Instructblip-Vicuna-7B model and an increase of 88.56 MME score points in the MME evaluation benchmark with the LLaVA-Next-8B model.

Comment: There needs to be an explicit research objective(s) stated, preferably as a separate section. This helps readers find out what the research is trying to address.

Response: We appreciate this suggestion. In response, we have added an explicit research objectives in the end of the Introduction to clearly state the objectives of this study. The revised content is provided below.

1.Introduction

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The LVLM utilizes an optimal shallow layer selected by the predictor to generate the denoised prediction distribution and multiplies it with that of the final layer, effectively filtering noise from shallow layers while preserving key signals in the final layer. Extensive experiments demonstrate that ALD² achieves stable performance improvements and significantly mitigates hallucinations of LVLMs.

The objective of our work is to investigate visual hallucinations in LVLMs from a layer-wise noise perspective and to develop an effective decoding strategy for mitigating such errors. Specifically, this work aims to analyze the impact of noisy representations in shallow decoding layers on token prediction stability, to design an adaptive denoising decoding framework that suppresses shallow-layer noise while preserving reliable semantic signals from the final layer, and to introduce a lightweight predictor that dynamically identifies the optimal layer for noise suppression during inference without modifying model parameters.

Our contributions are as follows:

Comment: I have questions about your dataset. So, you need to explicitly provide a description of your dataset(s) in the manuscript, which is currently lacking.

Response: Thank you for your comment regarding the dataset description. We have explicitly provided a detailed description of the dataset(s) used in our work in **Section 4.1**, including their sources, preprocessing procedures, and key statistics. This information was previously mentioned in the manuscript, and we have now clarified it to ensure it is clearly presented, addressing your concern.

4.1. Experimental Setup

Tasks. Our framework is evaluated across various tasks. We assess object hallucinations on POPE (Li et al., 2023c) and CHAIR (Rohrbach et al., 2018) metrics. As for CHAIR metrics, we randomly select 500 images from the MSCOCO 2017 (Lin et al., 2014) validation dataset. For the POPE metrics, we select 500 images from the MSCOCO 2014 (Lin et al., 2014) validation dataset and assign six queries to each image. For LVLMs' general ability, we employ MME (Fu et al., 2025) benchmark. These benchmarks are respected in the field for their diversity and breadth of tasks, allowing us to comprehensively analyze the performance of LVLMs from multiple perspectives. All these tasks are designed as open-ended generation tasks.

Comment: You must more clearly highlight your research's theoretical and practical implications; it is unclear how this research differs from existing work. There should be an explicit section addressing the discussion of results and implications.

Response: Thank you for your comment regarding the need to more clearly highlight the theoretical and practical implications of our research. In response, we have added a dedicated section (Section 6) in the

revised manuscript that explicitly discusses the implications of our findings and clarifies how our work differs from existing studies.

6. Theoretical and Practical Implications

Our work provides both theoretical and practical insights into mitigating visual hallucinations in large vision-language models. From a theoretical perspective, our findings suggest that hallucinations are not solely determined by the final decoding layer but are strongly influenced by noisy representations in shallow layers. By empirically demonstrating that correct tokens often receive relatively higher probabilities at shallow layers before being overridden during deep decoding, this work reframes hallucination as a layer-wise noise propagation problem rather than a purely semantic misalignment issue. This perspective complements and extends existing contrastive decoding approaches, which primarily focus on input perturbations, by highlighting the importance of internal layer dynamics during inference.

From a practical perspective, the proposed ALD² framework offers a lightweight solution for reducing hallucinations in LVLMs. ALD² operates entirely at the decoding stage and can be seamlessly integrated into existing LVLM pipelines. The adaptive layer selection mechanism further enables dynamic noise suppression tailored to different inputs, making the approach suitable for real-world deployment scenarios where reliability are critical. Together, these implications demonstrate that layer-wise denoising during decoding is both a theoretically grounded and practically effective strategy for improving the trustworthiness of LVLMs.

Comment: The manuscript requires thorough copy editing. Please keep in mind that many of the IP&M audience members are not native English speakers, so the manuscript must be accessible to them. Coherent English greatly aids in this.

Response: Thank you for your suggestion regarding the clarity and readability of the manuscript. We have carefully reviewed the language throughout the revised version and significantly improved the English to ensure it is more coherent and accessible to a broad audience, including non-native English speakers. We believe these revisions enhance the overall readability and clarity of the paper.

Comment: The manuscript is short, so you have space to go into more detail concerning this research’s prior literature, methods, data, and implications.

Response: Thank you for your constructive suggestion regarding the need for more detailed discussion of the prior literature, the methodology, the dataset, and the implications of our research. In response, we have expanded the corresponding parts of the manuscript below. We add one figure of the current method to mitigate the hallucinations in LVLMs as **Figure 4** and the details of metric calculations as **Appendix A**.

Comment: Your reference list needs tidying up, as there are references missing items or formatting issues. Please be consistent with the formatting and use some standard formatting style. IP&M uses APA for published articles if this helps.

Response: Thank you for your comment regarding the reference list formatting. We have carefully revised and tidied up the references to ensure that all entries are complete and consistently formatted according to a standard style. In particular, we have aligned the reference list with APA style, as recommended for Information Processing & Management, and corrected any missing items or formatting

issues to improve accuracy and readability.

Comment: Please review recent IP&M articles on similar topics to get some insights on manuscript structure, presentation, and level of research.

Response: Thank you for your suggestion to review recent Information Processing & Management articles on related topics for insights on manuscript structure, presentation, and research level. We have consulted the journal's recent publications and author guidelines to better align our revised manuscript with the journal's standards and expectations. In doing so, we paid attention to how articles in Information Processing & Management present research contributions, structure related work, and emphasize methodological and practical relevance. This has helped us refine the organization and presentation of our literature review, methods, and implications to improve clarity and fit the scope of the journal.

Comment: Please see IP&M's author guidelines, which may assist you in preparing your manuscript.

Response: Thank you for your suggestion to consult Information Processing & Management's author guidelines. We have carefully reviewed the journal's submission requirements and formatting instructions to ensure that our manuscript conforms to the journal's expectations. Specifically, we have followed the author guidance provided on the Information Processing & Management website, including requirements for manuscript structure, file formats, and other preparation details, so that the revised submission aligns with the journal's standards.